Tousik Samui, PhD

Physicist | Data Science, Machine Learning & Scientific Computing | HPC

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Profile

Physicist (PhD, TIFR) with eight years of postdoctoral research in particle physics, built on large-scale simulation, statistics, machine learning and software. I write well-tested C++ and Python, run reproducible pipelines on terabyte-scale data, and have administered a 440-core HPC cluster and built its live monitoring dashboard. Author of DynamicR, an open-source package in FastJet Contrib; 18 peer-reviewed papers (10 as corresponding author) and 40 talks. Open to roles in data science, machine learning, quantitative research, HPC and research software engineering, as well as academia.

Core skills

Programming	C++ (object-oriented design, STL, plugins for large libraries), Python (NumPy, pandas, scikit-learn, XGBoost, Matplotlib), C, Fortran, Bash, PHP, HTML/CSS, JavaScript (jQuery), Mathematica; Git/GitHub
Data science & ML	Supervised classification (boosted decision trees, multivariate analysis), unsupervised learning (self-organising maps), feature engineering on high-dimensional data, imbalanced data, model evaluation (ROC/AUC, significance)
Statistics & modelling	Probability, likelihood-based inference, hypothesis testing and multiple-testing control, regression, Monte Carlo simulation, unfolding (inverse problems), uncertainty estimation
HPC & systems	Linux administration (RHEL / AlmaLinux), PBS and Slurm schedulers, cluster operation, software builds, monitoring dashboards, user support, terabyte-scale data workflows
Scientific software	ROOT, FastJet, Pythia8, Delphes, MadGraph5, CalcHEP, FeynRules, SARAH, SPheno, MicrOmegas
Communication	Technical writing (18 papers), 40 talks, teaching, documentation, mentoring PhD students

Experience

Post Doctoral Fellow — Jun 2024 – present
The Institute of Mathematical Sciences (IMSc) — *Chennai, India*

- Lead research projects end to end with PhD students and collaborators at other institutes: problem definition, simulation, analysis, statistical interpretation and publication.
- Build automated, reproducible simulation and analysis pipelines in C++, Python and Bash that process terabyte-scale data sets on the institute HPC cluster.
- Applied the Dynamic Radius clustering method to public CMS Open Data in a search for new heavy particles, with a clear gain in sensitivity (Phys. Rev. D 2026); invited to present it to the CMS B2G group at CERN (2026).
- Developed an unfolding method (arXiv:2512.09121) that corrects detector distortions in measured quantum-entanglement observables.
- Released DynamicR into FastJet Contrib (v1.104) and generalised the algorithm (arXiv:2609.29862).

Key skills: C++, Python, Bash, statistical inference, unfolding, data pipelines, HPC, open-source release, project leadership

Post-doctoral Fellow — Nov 2021 – May 2024
Indian Institute of Science Education and Research (IISER) Kolkata — *Kolkata, India*

- Co-invented the Dynamic Radius jet clustering algorithm, in which each cluster's size adapts to the local data (JHEP 2023), and wrote its C++ implementation with documentation, an example and a regression test.
- Built multivariate classifiers (boosted decision trees) on high-dimensional jet features and evaluated them with ROC curves and significance, e.g. W-boson polarisation (Eur. Phys. J. C 2023).
- Published seven peer-reviewed papers; organised the weekly HEP journal club; teaching assistant for Mechanics I; Secretary of the MODE collaboration (2023–25).

Key skills: algorithm design, C++, machine-learning classification, feature engineering, model evaluation, technical writing

System Administrator and Maintenance Coordinator (additional role) — Apr 2023 – May 2024
Departmental HPC cluster "Kepler", IISER Kolkata — *Kolkata, India*

- Ran the shared cluster of the Department of Physical Sciences: 11 compute nodes (440 CPU cores, one GPU node, about 2 TB memory), PBS scheduler, AlmaLinux. Managed accounts, queues, software installation, networking and resource planning.
- First point of contact for users: fixed failed jobs, build errors and missing libraries, and turned common problems into short guides.
- Designed and built the cluster website and a live dashboard of node states, free CPUs, per-user CPU efficiency and storage use (PHP, HTML/CSS, JavaScript/jQuery); still in use in 2026: iiserkol.ac.in/~dps.hpc/clusterStatus.

Key skills: Linux administration, PBS, Bash, PHP, JavaScript, monitoring, user support, documentation

Post-Doctoral Fellow Sep 2018 – Sep 2021
Harish-Chandra Research Institute (HRI) *Prayagraj, India*
Three-year fixed-term contract; I stayed until Nov 2021 under an unpaid COVID-19 extension.

- Research on physics beyond the Standard Model and jet substructure; six peer-reviewed papers.
- Ran large Monte Carlo campaigns on the institute cluster and used jet-substructure features to classify new heavy particles (Eur. Phys. J. C 2021).
- Designed and maintained a live job-status web page for the HRI HPC cluster (2020–21).
- Organised the weekly HEP journal club; coordinated and taught at Harsha School, an outreach programme for underprivileged school students.

Key skills: Monte Carlo simulation, classification, data analysis, PHP/HTML, outreach

Doctoral researcher (PhD student) Aug 2012 – Sep 2018
Tata Institute of Fundamental Research (TIFR) *Mumbai, India*

- Wrote an in-house Monte Carlo event generator for a near-continuous spectrum of new particle states that standard tools could not handle; thesis defended in 2019.
- Teaching assistant for three graduate courses; research visits to LPT Orsay, SNS Pisa and LPSC Grenoble, and the Les Houches workshop (2017).

Key skills: Monte Carlo methods, numerical simulation, analytical modelling

Education

PhD in Physics, Tata Institute of Fundamental Research, Mumbai 2019
MSc in Physics, Indian Institute of Technology Kanpur 2012
BSc (Hons) in Physics, Ramakrishna Mission Residential College, Narendrapur (University of Calcutta) 2010

Awards and national ranks

- All India Rank **3**, CSIR-NET Physical Sciences (2013); AIR **43**, CSIR-NET Mathematical Sciences (2012).
- All India Rank **15**, GATE Mathematics (2013); AIR **23**, GATE Physics (2011); AIR **5**, JAM Physics (2010); AIR **16**, JEST (2011).
- Best Poster, DPS Day 2023 (IISER Kolkata); Academic Excellence Award, MSc Physics (IIT Kanpur, 2011).
- CSIR Shyama Prasad Mukherjee Fellowship (2013); DST INSPIRE Scholarship (2007–12).

Publications and talks

18 peer-reviewed papers (JHEP, Phys. Rev. D, Eur. Phys. J. C, Phys. Lett. B, Nucl. Phys. B), 10 as corresponding author; 4 preprints; 2 proceedings; 40 talks, including invited talks to the CMS B2G group (2026) and the ATLAS L2IT group (2024). Full list: tousiksamui.github.io/pub.html. Selected (* corresponding author; authors in alphabetical order):

- B. Mukhopadhyaya, **T. Samui***, R. K. Singh, *Dynamic Radius Jet Clustering Algorithm*, JHEP 04 (2023) 019.
- A. Ghosh, S. Ghosh, S. Mitra, **T. Samui***, R. K. Singh, *Improving Sensitivity of Vectorlike Top Partner Searches with Jet Substructure*, Phys. Rev. D 113 (2026) 035034.
- A. Dey, **T. Samui***, *Polarization study of a boosted W boson decaying hadronically at the LHC with jet substructures and a multivariate analysis*, Eur. Phys. J. C 83 (2023) 1002.
- A. K. Nayak, S. K. Rai, **T. Samui***, *Improving Heavy Dijet Resonance Searches Using Jet Substructure at the LHC*, Eur. Phys. J. C 81 (2021) 130.

Teaching, leadership and outreach

- Lectures: *Data handling using ROOT*, ML4HEP school, IISER Kolkata (2025); *Introduction and quantization of fields*, PWF Bangladesh school on QFT with ICTP (2024); *Linear representations of groups*, recorded series (2022).
- Teaching assistant: Classical Mechanics, Quantum Field Theory I, Electrodynamics II (TIFR); Mechanics I (IISER Kolkata).
- Secretary, MODE collaboration (machine-learning-optimised design of experiments), 2023–25; journal-club organiser at HRI and IISER Kolkata; Harsha School outreach (2019–21).

Languages: English (fluent), Bengali (native), Hindi. **Profiles:** INSPIRE • ORCID 0000-0002-1485-6155 • arXiv. References available on request.